

Cross-Track Integration Report

HealthConnect — Patient Attendance & No-Show Analysis

Intern: [Wisdom Chibuike Ukah]

Track: Data Analytics

Reporting Period: Week 6

1. Track Collaborated With

Data Science Track

Collaborator: [Dorsilla Kemunto]

Artifacts exchanged: Week 5 & Week 6 dashboards (Analytics) ↔ Week 6 model outputs (Data Science: target/feature setup, error distribution, Random Forest feature importance).

2. Project Dependency

The **Data Analytics** track produces the business-facing segmentation and insight layer (which patients miss appointments, why, and through which channels). The **Data Science** track produces the predictive layer (a model that flags likely no-shows before the appointment).

These are directly dependent:

- **Analytics → Data Science:** The analytical dimensions I identified (booking lead time, distance to clinic, appointment type, age group, reminder channel) are the candidate features the model needs to be useful.
- **Data Science → Analytics:** The model's feature importance ranks confirm or challenge which analytical findings have the strongest predictive signal, which shapes the recommendations I make to HealthConnect.

Without this exchange: Analytics recommendations would remain descriptive with no validation, and the Data Science model would keep training on an impoverished 4-feature set that does not include the drivers my analysis identified.

3. Information Received

From the Data Science track, I received:

Item	Content
Target & feature setup	Target = no_show; Week 5 ML-ready dataset contained only 4 features (Unnamed: 0, previous_no_shows, waiting_time_minutes, appointment_day)
Feature importance	Random Forest ranking — #1 numeric_booking_lead_days, #2 numeric_distance_to_clinic_km, #3 numeric_Unnamed: 0, #4 numeric_age, #5 numeric_previous_no_shows
Error distribution	TN = 259 (27.32%), TP = 243 (25.63%), FN = 242 (25.53%), FP = 204 (21.52%) — ~53% accuracy
Model class	Random Forest (upgraded from Week 5 model)

4. Information Provided

To the Data Science track, I provided:

Item	Content
Week 5 dashboard	Baseline KPI validation, no-show rate by reminder channel (WhatsApp best at 49.8% no-show)
Week 6 dashboard	Attendance rate by reminder channel, no-show rate by distance × age group, highest-risk combinations (distance × lead time), attendance by lead time × appointment type
Derived segmentation dimensions	Distance buckets (Close / Moderate / Far / Very Far), lead-time buckets (Short Term / 1–2 Weeks / 2–4 Weeks / 1–2 Months), risk-combination flags
Analytical findings	Top 6 highest-risk combinations; confirmation that booking lead time and distance are the two dominant drivers

5. Integration Activity Completed

I performed a **cross-method triangulation** between my manual analytical segmentation (Analytics) and the model's feature importance ranking (Data Science). This is a deliberate analytical activity — not a conversation — with three concrete outputs:

A. Driver convergence test. I mapped my Week 6 analytical findings against the model's top-ranked features to see whether two independent methods identified the same drivers.

Week 6 Analytical Insight	ML Feature Importance Rank	Convergence
Booking lead time drives attendance (Section 3)	#1 — numeric_booking_lead_days	✓ Strong
Distance to clinic drives no-shows (Sections 4 & 5)	#2 — numeric_distance_to_clinic_km	✓ Strong
Age is a risk modifier (Section 4)	#4 — numeric_age	✓ Match
Prior no-shows matter	#5 — numeric_previous_no_shows	✓ Match
Reminder channel affects attendance (Section 2)	#10–11 — reminder_sent_Yes/No	⚠ Weak signal

6. What Changed as a Result

The collaboration produced **four measurable changes** — not just an exchange of information:

#	Change	Type
1	Analytical findings were independently validated. Two unrelated methods (manual segmentation and Random Forest feature importance) converged on the same top two drivers. This upgrades my Week 6 recommendations from "observed pattern" to "cross-validated finding."	Validated output
2	A model defect was identified and reported. numeric_Unnamed: 0 was flagged as an index leak in the #3 feature slot — a data-quality issue that, if left unchecked, inflates apparent model performance and misleads feature interpretation.	Improvement
3	The Data Science feature set was redirected. I provided the Week 6 segmentation dimensions (distance buckets, lead-time buckets, reminder channel, appointment type × lead time) as candidate features for the next model iteration — expanding it beyond the current 4-feature limit.	Integration
4	The model's success metric was redefined. I recommended shifting from overall accuracy (~53%) to recall on the no-show class, because False Negatives carry the highest business cost in a scheduling context.	Decision
Net result: My Week 6 report gained a validated, ML-backed evidence layer; the Data Science model gained a richer feature roadmap and a corrected evaluation target.		

7. Evidence

Evidence Type	Artifact
Analytics artifacts shared	Week 5 dashboard (KPI validation + no-show rate by reminder channel); Week 6 dashboard (4 visuals + risk combination tables)
Data Science artifacts received	Screenshots of target/feature setup code, Random Forest feature importance chart, error distribution table
Communication trail	[https://analystlabafrika.slack.com/archives/D0C0GM9JERK]

Summary Statement

The Analytics ↔ Data Science collaboration was not limited to communication. It produced a **validated cross-method finding** (booking lead time and distance confirmed as the top two no-show drivers by both manual analysis and Random Forest feature importance), a **data-quality correction** (index leak in the model), a **feature roadmap** (Week 6 dimensions handed to Data Science), and a **metric redefinition** (recall over accuracy). These outputs changed both the Week 6 analytical report and the direction of the next model iteration.